

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 3– Output Prediction Questions

Course Code:	CSA0315	Course Title:	Data Structures
CO Assessed:	CO1	Assessment:	Assessment Tool 3 – Debugging and Optimization Questions
Weightage:	10% of CIA	Total Marks:	100
CO2 Statement:	Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.		
Student Name:	Peram.Mahitha	Reg. No.:	192524136
Section / Batch:	2025	Date:	29-07-2026

Code of Conduct

I Peram.Mahitha(192524136) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ P.Mahitha _____

Instructions

- This test contains 10 MCQ Based Questions. Total: 100 Marks.
- When a student answer using saveetha compiler, execute the code, Test the test cases , While evaluation the answer should contain following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- Electronic Gadgets are permitted in the class. Take it as test and submit in the saveetha lab compiler class.

Annexure A – Output Prediction Questions

1. Write a C program that takes an array as input and returns it reversed.
2. Write a C program for palindrome
3. Write a C program for Searching an Element in a Sorted Array
4. Write a C program for Count the Occurrences of an Element
5. Write a program for binary search
6. Write a C program for Merge Two Arrays
7. Write a C program for Find the Largest and Smallest Element in an Array
8. Write a C program for Find the Second Largest Element
9. Write a C program for Find the First and Last Occurrence of an Element
10. Write a C program for Find Missing Element in an Array

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

Q4. Searching

CODE

```
#include<stdio.h>
int main(){
    int n,i,key,count=0;
    scanf("%d",&n);
    int a[n];
    for(i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    scanf("%d",&key);
    for(i=0;i<n;i++){
        if(a[i]==key){
            count++;
        }
    }
    printf("%d",count);
    return 0;
}
```

INPUT

```
7
2 4 5 4 6 4 8
4
```

OUTPUT

```
3
```

Report

Q5. Array

CODE

```
#include<stdio.h>
int main(){
    int n,i,key;
    int low,high,mid;
    scanf("%d",&n);
    int a[n];
    for(i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    scanf("%d",&key);
    low=0;
    high=n-1;
    while(low<=high){
        mid=(low+high)/2;
        if(a[mid]==key){
            printf("element found at position %d",mid+1);
            return 0;
        }
        else if(a[mid]<key)
        {
            low=mid+1;
        }
        else{
            high=mid-1;
        }
    }
    printf("element not found");
    return 0;
}
```

INPUT

```
5
10 20 30 40 50
30
```

Q9. Occurrence of an elemnet

CODE

```
#include<stdio.h>
int main(){
    int n,i,key;
    int first=-1,last=-1;
    scanf("%d",&n);
    int a[n];
    for(i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    scanf("%d",&key);
    for(i=0;i<n;i++){
        if(a[i] == key){
            if(first == -1)
                first=i;
            last=i;
        }
    }
    if(first == -1)
        printf("Element not found");
    else
    {
        printf("first occurence=%d\n",first);
        printf("last occurence=%d\n",last);
    }
    return 0;
}
```

INPUT

```
7
2 4 5 4 6 4 8
4
```

OUTPUT

(no output)

Q10. Missing Element

CODE

```
#include<stdio.h>
int main(){
    int i,n,sum=0,total;
    scanf("%d",&n);
    int a[n-1];
    for(i=0;i<n-1;i++){
        scanf("%d",&a[i]);
        sum=sum+a[i];
    }
    total=n*(n+1)/2;
    printf("missing element=%d",total-sum);
    return 0;
}
```

INPUT

```
5
1 2 3 5
```

OUTPUT

(no output)

